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Number System

FUNDAMENTAL CONCEPTS

I. Numbers

In Hindu-Arabic system, we have ten digits, namely 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

A number is denoted by a group of digits, called **numeral**.

For denoting a numeral, we use the place-value chart, given below.

	Ten-Crores	Crores	Ten-Lakhs	Lakhs	Ten-Thousands	Thousands	Hundreds	Tens	Ones
(i)				5	2	8	6	7	9
(ii)			4	3	8	0	9	6	7
(iii)		3	5	2	1	8	0	0	9
(iv)	5	6	1	3	0	7	0	9	0

The four numerals shown above may be written in words as:

- (i) Five lakh twenty-eight thousand six hundred seventy-nine
- (ii) Forty-three lakh eighty thousand nine hundred sixty-seven
- (iii) Three crore fifty-two lakh eighteen thousand nine
- (iv) Fifty-six crore thirteen lakh seven thousand ninety

Now, suppose we are given the following four numerals in words:

- (i) Nine crore four lakh six thousand two
- (ii) Twelve crore seven lakh nine thousand two hundred seven
- (iii) Four lakh four thousand forty
- (iv) Twenty-one crore sixty lakh five thousand fourteen

Then, using the place-value chart, these may be written in figures as under:

	Ten-Crores	Crores	Ten-Lakhs	Lakhs	Ten-Thousands	Thousands	Hundreds	Tens	Ones
(i)		9	0	4	0	6	0	0	2
(ii)	1	2	0	7	0	9	2	0	7
(iii)				4	0	4	0	4	0
(iv)	2	1	6	0	0	5	0	1	4

II. Face value and Place value (or Local Value) of a Digit in a Numeral

- (i) The face value of a digit in a numeral is its own value, at whatever place it may be.

Ex. In the numeral 6872, the face value of 2 is 2, the face value of 7 is 7, the face value of 8 is 8 and the face value of 6 is 6.

- (ii) In a given numeral:

Place value of ones digit = (ones digit) $\times$ 1,

Place value of tens digit = (tens digit) $\times$ 10,

Place value of hundreds digit = (hundreds digit) $\times$ 100 and so on.

Ex. In the numeral 70984, we have

Place value of 4 = $(4 \times 1) = 4$,

Place value of 8 = $(8 \times 10) = 80$,

Place value of 9 = $(9 \times 100) = 900$,

Place value of 7 = $(7 \times 10000) = 70000$.

Note: Place value of 0 in a given numeral is 0, at whatever place it may be.

III. Various Types of Numbers

1. **Natural Numbers:** Counting numbers are called natural numbers.

Thus, 1, 2, 3, 4, are all natural numbers.

2. **Whole Numbers:** All counting numbers, together with 0, form the set of whole numbers.

Thus, 0, 1, 2, 3, 4, are all whole numbers.